

Xinkai Wang, Postdoc Researcher

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🏠 No. 800 Dongchuan Road, Minhang District, Shanghai, China, 200240
📖 **Research Area: Computer Architecture, Computer System**



Research Interests

My research area is computer architecture and systems, with a primary focus on efficient and reliable datacenter management optimization, including:

- 📌 **High Energy Efficiency:** How to better model, allocate, and schedule heterogeneous and constrained hardware resources to improve system performance per watt?
- 📌 **High Resilience:** How to better analyze, detect, and eliminate frequent hardware and software failures to ensure stable system operation?

Currently, I mainly focus on the following two research topics:

- 📌 **Energy-efficient Design Exploration for Agentic AI Infrastructure**
- 📌 **Agentic Management for High-resilience Infrastructure**

Education/Work Experiences

- 📌 **Postdoc Researcher**, Shanghai Jiao Tong University. 2026.07 – now
Advisor: **Prof. Minyi Guo**
- 📌 **Ph.D.**, Computer Science, Shanghai Jiao Tong University. 2021.09 – 2026.06
Advisor: **Prof. Chao Li**
Thesis: *Optimizing datacenter manageability for low interference and high efficiency*
Outstanding Graduate
- 📌 **Visiting Scholar**, Computer Science, Ghent University, Belgium. 2025.10 – 2026.03
Advisor: **Prof. Lieven Eeckhout**, ELIS Department
- 📌 **B.Engr.**, Computer Science, Shanghai Jiao Tong University. 2017.09 – 2021.06
Thesis: *Exploring service level parallelism in cloud native era* (Outstanding Thesis)
Outstanding Graduate, Zhiyuan Honors Program of Engineering

Research Publications

Overall, I have published 18 papers, including 8 first-author papers (4 CCF-A, 3 CCF-B). I have filed 11 patents, including 6 granted patents. Also, I have 2 first-author CCF-A papers in submission.

Conference Proceedings

- 1 **Xinkai Wang**, Chao Li, Yiming Zhuansun, Jinyang Guo, Xiaofeng Hou, Jing Wang, et al., “Aum: Unleashing the efficiency potential of shared processors with accelerator units for llm serving,” in *Proc. the 32nd IEEE International Symposium on High-Performance Computer Architecture (HPCA 2026)*, CCF-A, 119/602=19.7%. 🔗 DOI: 10.1109/HPCA68181.2026.11408539
- 2 **Xinkai Wang**, Xiaofeng Hou, Chao Li, Yuancheng Li, Du Liu, Guoyao Xu, et al., “Exist: Enabling extremely efficient intra-service tracing observability in datacenters,” in *Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Volume 2 (ASPLOS 2025)*, CCF-A, 160/912=17.5%, pp. 355–372. 🔗 DOI: 10.1145/3676641.3716283
- 3 **Xinkai Wang**, Yiming Zhuansun, Chao Li, Jing Wang, Xiaofeng Hou, Lingyu Sun, et al., “Asymserve: Demystifying and optimizing llm serving efficiency on cpu acceleration units,” in *Advanced Parallel Processing Technologies (APPT 2025)*, CCF-C, 17/74=22.9%, pp. 231–245. 🔗 DOI: 10.1007/978-981-95-1021-4_17

- 4 **Xinkai Wang**, Chao Li, Lingyu Sun, Qizheng Lyu, Xiaofeng Hou, Jingwen Leng, et al., “Sheeo: Continuous energy efficiency optimization in autonomous embedded systems,” in *2024 IEEE 42nd International Conference on Computer Design (ICCD 2024, CCF-B, Best Paper Nomination*, 74/277=26.7%), pp. 496–503.  DOI: 10.1109/ICCD63220.2024.00082
- 5 **Xinkai Wang**, Hao He, Yuancheng Li, Chao Li, Xiaofeng Hou, Jing Wang, et al., “Not all resources are visible: Exploiting fragmented shadow resources in shared-state scheduler architecture,” in *Proceedings of the 2023 ACM Symposium on Cloud Computing (SoCC 2023, CCF-B, 40/133=30%)*, pp. 109–124.  DOI: 10.1145/3620678.3624650
- 6 **Xinkai Wang**, Chao Li, Lu Zhang, Xiaofeng Hou, Quan Chen, and Minyi Guo, “Exploring efficient microservice level parallelism,” in *2022 IEEE International Parallel and Distributed Processing Symposium (IPDPS 2022, CCF-B, 123/474=25.9%)*, pp. 223–233.  DOI: 10.1109/IPDPS53621.2022.00030
- 7 Chang Liu, Xiaofeng Hou, Cheng Xu, **Xinkai Wang**, Jiacheng Liu, Chao Li, et al., “Locmore: Locating more bursty latency-critical jobs on resource-constrained nodes,” in *IEEE/ACM International Symposium on Quality of Service (IWQoS 2026, CCF-B)*.
- 8 Mingxuan Zhang, Xiaofeng Hou, Jiacheng Liu, Chang Liu, **Xinkai Wang**, Chao Li, et al., “Starkserve: A framework for elastic serverless llm inference at the extreme edge,” in *IEEE/ACM International Symposium on Quality of Service (IWQoS 2026, CCF-B)*.
- 9 Jinyang Guo, **Xinkai Wang**, Jing Wang, Xiaofeng Hou, Chao Li, and Minyi Guo, “Tricooling-sim: Efficient thermal simulation for high-density micro ai data centers,” in *Network and Parallel Computing (NPC 2025, CCF-C, Best Student Paper)*, pp. 227–239.  DOI: 10.1007/978-3-032-10466-3_19
- 10 Jing Wang, Taolei Wang, Juntao Huang, Yibo Liu, **Xinkai Wang**, Marius Kreutzer, et al., “Accelerating large-scale out-of-gpu-core gnn training with two-level historical caching,” in *International Symposium on Advanced Parallel Processing Technology (APPT 2025, CCF-C, Distinguished Artifact Award)*.  DOI: 10.1007/978-981-95-1021-4_22
- 11 Lingyu Sun, Chao Li, Xiaofeng Hou, Tianhao Huang, Cheng Xu, **Xinkai Wang**, et al., “Jigsaw: Taming bev-centric perception on dual-soc for autonomous driving,” in *2024 IEEE Real-Time Systems Symposium (RTSS 2024, CCF-A)*, pp. 280–293.  DOI: 10.1109/RTSS62706.2024.00032
- 12 Du Liu, Jing Wang, **Xinkai Wang**, Chao Li, Lu Zhang, Xiaofeng Hou, et al., “Improving the efficiency of serverless computing via core-level power management,” in *2024 IEEE 24th International Symposium on Cluster, Cloud and Internet Computing (CCGrid 2024, CCF-C)*, pp. 125–135.  DOI: 10.1109/CCGrid59990.2024.00024
- 13 Lu Zhang, Chao Li, **Xinkai Wang**, Weiqi Feng, Zheng Yu, Quan Chen, et al., “First: Exploiting the multi-dimensional attributes of functions for power-aware serverless computing,” in *2023 IEEE International Parallel and Distributed Processing Symposium (IPDPS 2023, CCF-B)*, pp. 864–874.  DOI: 10.1109/IPDPS54959.2023.00091

Journal Articles

- 1 **Xinkai Wang**, Chao Li, Tianhao Huang, Jinyang Guo, Xiaofeng Hou, Guoyao Xu, et al., “Intra-service tracing for shared and stressed datacenters: efficient management and intelligent analysis,” *Science China Information Sciences (SCIS, CCF-A, Chinese, Impact Factor=7.6)*, 2026.
- 2 **Xinkai Wang**, Chao Li, Yiyang Li, Lingyu Sun, Cheng Xu, Xiaofeng Hou, et al., “Enabling learning-based efficiency optimizer with shadow cycles in resource-constrained autonomous embedded systems,” *IEEE Transactions on Computers (TC, CCF-A, Impact Factor=3.8)*, vol. 75, no. 3, pp. 928–941, 2025.  DOI: 10.1109/TC.2025.3644184
- 3 Junyi Mei, Shixuan Sun, Chao Li, **Xinkai Wang**, Jing Wang, Xiaofeng Hou, et al., “Dgs: A gpu-based adaptive graph sampling framework,” *ACM Transactions on Architecture and Code Optimization (TACO, CCF-A)*,  DOI: 10.1145/3817060

- 4 Du Liu, Lu Zhang, Yechen Xu, **Xinkai Wang**, Lingyu Sun, Yifei Pu, et al., “Power synchronization: Taming massive diversified serverless functions under power constraints,” *Science China Information Sciences (SCIS, CCF-A)*, vol. 68, no. 132101, 2025.  DOI: 10.1007/s11432-022-3882-y
- 5 Lingyu Sun, Xiaofeng Hou, Chao Li, Jiacheng Liu, **Xinkai Wang**, Quan Chen, et al., “A2: Towards accelerator level parallelism for autonomous micromobility systems,” *ACM Transactions on Architecture and Code Optimization (TACO, CCF-A)*, vol. 21, no. 4, 2024.  DOI: 10.1145/3688611

Granted Patents

- 1 Chao Li, Lu Zhang, Yechen Xu, **Xinkai Wang**, Jingwen Leng, Quan Chen, et al., Chinese Patent Granted, CN115129475B, May 12, 2026.
- 2 Chao Li, Hanzhang Yang, Jing Wang, **Xinkai Wang**, and Minyi Guo, Chinese Patent Granted, CN119728775B, 2025.
- 3 Chao Li, Lingyu Sun, **Xinkai Wang**, and Minyi Guo, Chinese Patent Granted, CN116842994B, 2024.
- 4 Chao Li, **Xinkai Wang**, Lingyu Sun, and Qizheng Lv, Chinese Patent Granted, CN116414556B, 2024.
- 5 Chao Li, **Xinkai Wang**, Lu Zhang, Zhexuan Chen, Quan Chen, and Minyi Guo, Chinese Patent Granted, CN114205419B, 2023.
- 6 Chao Li, Lu Zhang, Weiqi Feng, Zheng Yu, **Xinkai Wang**, and Minyi Guo, Chinese Patent Granted, CN113238853B, 2021.

Honors and Awards

Paper Awards		Best Student Paper Award , IFIP NPC 2025, (2/223)	Nov. 2025
		Distinguished Artifact Award , APPT 2025, (2/74)	Jul. 2025
		Best Paper Nomination , ICCD 2024, (4/277)	Nov. 2024
Academic Honors		Best Presentation Award , Ph.D. Forum at CCF Computility 2025	Jul. 2025
		Champion Award , Graduate Track of CCF TCARCH SRC	Jul. 2025
		Travel Grants : ISCA 2025, ASPLOS 2025, SoCC 2023, IPDPS 2023	
Other Honors		Outstanding Student Cadres , Shanghai Jiao Tong University	Oct. 2022
		Outstanding League Cadres , Shanghai Jiao Tong University	May 2022
		Yang Yuan Qing Excellent Scholarship , SJTU CSE Department	May 2021
		Zhiyuan Honor Scholarship , Shanghai Jiao Tong University	2017-2021

Project Experience

Government Funded Projects

-  **Agile Design of Open-Source General Processors** (No.U23A6007) 2024-2027
Participant, NSFC Joint Fund for Regional Innovation and Development
-  **System Software for Micro Data Centers** (No. 2022YFB4501702) 2023-2026
Participant, National Key Research and Development Program of China
-  **Resource Management for Scalable Data Centers** (No. 62122053) 2022-2024
Participant, NSFC Excellent Young Scientists Fund

Industry Funded Projects

-  **Modeling and Analysis of CPU-centric Agentic Infrastructure** 2026-2027
Team Leader, AIS, Alibaba Group

Project Experience (continued)

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| ■ | Efficient Hardware Fault Tolerance in AI Infrastructure | 2026-2027 |
| | Team Leader, TRE, Alibaba Group | |
| | Technology is submitted to ASPLOS 2027 as first author. | |
| ■ | Performance Modeling of Datacenter Shared Workloads for CPU Design | 2025-2026 |
| | Team Leader, AIS, Alibaba Group | |
| | Technology is submitted to ASPLOS 2027 as co-first author. | |
| ■ | HW-SW Co-Optimization Technologies for Large-Scale AI Infrastructure | 2024-2025 |
| | Team Leader, TRE, Alibaba Group | |
| | Technology is published on HPCA 2026 [1] and deployed in large-scale datacenters. | |
| ■ | Performance Bottleneck Diagnosis for Cloud-Native Infrastructure | 2023-2024 |
| | Research Intern, TRE, Alibaba Group | |
| | Technology is published on ASPLOS 2025 [2] and deployed in large-scale datacenters. | |

Professional Activities

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| Reviewer | ■ | <i>Artifact Evaluation Committee</i> of ISCA 2026, HPCA 2026, ASPLOS 2025-2026, OSDI 2025, ATC 2025
<i>Shadow PC</i> of EuroSys 2024
IEEE Transactions on Reliability (TR)
IEEE Transactions on Sustainable Computing (TSUSC) | |
| Services | ■ | <i>Volunteer</i> at APPT 2025-2026, ASPLOS 2025, CCF CHIPS 2024, IPDPS 2023, etc.
<i>Translator</i> of CCCF (2025 1(3), 2022 18(12))
<i>DRA</i> of IEEE C/AISC Standard P3584
<i>Tutor</i> of the undergraduate of CSE department. | 2026
2021-2025 |
| Teaching | ■ | TA of Eo34140-Parallel Computing Systems, Ghent University
TA of CS3328-Cloud Computing Technology, SJTU
TA of CS359-Computer Architecture, SJTU | Fall, 2025
Fall, 2021-2022
Fall, 2019 |
| Membership | ■ | Student member of China Computer Federation (CCF)
Student member of Association for Computing Machinery (ACM)
Student member of Institute of Electrical and Electronics Engineers (IEEE) | |

Talks

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| ■ | Unleashing the Efficiency Potential of Shared Processors with Accelerator Units for LLM Serving | |
| | ChinaSys 2026, Xi'an | 2026 |
| | Conference Talk, HPCA 2026, Sydney, Australia | 2026 |
| | APPT 2025 Student Research Challenge, Online | 2025 |
| ■ | Enabling Extremely Efficient Intra-Service Tracing Observability in Datacenters | |
| | CCF SYS 2025, Beijing | 2025 |
| | Conference Talk, ASPLOS 2025, Rotterdam, The Netherlands | 2025 |
| ■ | Continuous Energy Efficiency Optimization for Autonomous Embedded Systems | |
| | CCF Computility 2025, Lanzhou | 2025 |
| | Intelligent Automotive Solution BU Seminar, Huawei, Shanghai | 2024 |
| | Conference Talk, ICCD 2024, Milan, Italy | 2024 |
| ■ | Exploiting Fragmented Shadow Resources in Shared-State Scheduler Architecture | |
| | Conference Talk, SoCC 2023, Santa Cruz, USA | 2023 |
| ■ | Exploring Efficient Microservice Level Parallelism | |
| | Conference Talk, IPDPS 2022, Online | 2022 |

Mentoring

Undergrad	📖	Ziqi Chen (BS'2025 @ Purdue, SDC mitigation of LLM serving) Yiyang Li (BS'2022 → MS @ CUHK, TC [15]) Qizheng Lyu (BS'2020 → MS @ SJTU, ICCD 2024 [4])
Master	📖	Qirong Du (MS'2026, CPU-centric Agentic Infrastructure) Yunwei Li (MS'2024, Efficient GPU fault tolerance) Yiming Zhuansun (MS'2023 → Ubiquant Investment, APPT 2025 [3], HPCA 2026 [1]) Yuancheng Li (MS'2021 → Alibaba, ASPLOS 2025 [2], SoCC 2023 [5])
Ph.D.	📖	Tianhao Huang (PhD'2025, Efficient cloud performance modeling) Lingyu Sun (PhD'2024, TACO 2024 [18], RTSS 2024 [11])